Human Resource Planning in the Tech Industry

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Lecture Outline

- What is HRP?
- Importance of HRP
- Process of HRP
- Forecasting future demand for HR
- Estimating HR Supply
- Compare demand with estimated supply
- Develop Strategies.

What is Human Resource Planning?

HRP means the forecasting the future employee needs and determining the steps or strategies to achieve those needs to the purpose of accomplishing organization goals and objectives .

Common purpose of the HRP is preparation of the organization to acquire the **appropriate employees** on the **appropriate place** at the **appropriate time** so as to accomplish the overall organization goals

Importance of HRP

1. To determine the future employee needs

2. To utilize the HR more efficiency and effectively

3. To control employee cost

4. To develop high talented employees

5. To formulate and implement strategic plans

HRP Process

Forecast future HR Demand

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Estimate HR Supply

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Compare HR demand with estimated

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Decides strategies to be taken

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Evaluate the process

01. Forecasting Demand for the HR

Factors affecting future HR demand

1. Demand for the product
2. Strategic goals of the Organization
3. Production Method
4. Retirements
5. Resignations
6. Deaths
7. Leave of absence
8. Termination
9. External factors such as economical, political and social factors

Talent Demand in Tech

- The Growing Need for Tech Talent:
  - Increasing reliance on technology across every sector: finance, healthcare, education, entertainment, etc.
  - Skills gap: Many emerging technologies (AI, blockchain, etc.) have led to a shortage of skilled workers who can fill these niche roles.

What are the key talent areas in IT?

- **Software Development**: Coding, application development, and software engineering.
- **Data Science and AI**: Machine learning, big data analytics, neural networks, etc.
- **Cybersecurity**: Protecting systems and networks from cyberattacks.
- **Cloud Computing**: Building and managing scalable, on-demand services in the cloud.
- **IT Support and Infrastructure**: Ensuring hardware and network stability and managing troubleshooting for tech issues.

The methods of forecasting

- **Informal Forecasting**

Easiest way to forecast the future human resource demand. Under this method requirement of future human resource forecasting without systematic basis.

It is done randomly and quickly.

Accuracy is low, generally good for small organization. This can not be recommended for the large scale organizations.

- **Formal Expert Survey**

The future human Resource demand is estimated by conducting a survey among the experts.

Using the questionnaire or interviews help to collect the information.

Long time and it takes higher degree accuracy.

High cost.

- **Delphi Techniques**

More sophisticated than the formal and informal method.

Usually the group of experts, HR Managers and HR department planners act as intermediary.

Independent individuals who have a greater knowledge in the field of planning, forecast the organizational HR demand and ultimately get aggregate average value of all forecasted figures.

- **Unit forecasting**

Broad approach, alternatively called as “Bottom up approach”

First each unit head of the organization has to do the personnel forecast with regard to the relevant unit.

The each head is supposed to forecast to his/her superior who is supposed to study forecast submitted by the all unit heads.

Finally head of the organization prepare the personnel forecasts for the entire organization

- **Extrapolation**

It is based on the past data and an attempt is made to find a trend or a pattern that might continue into the future. Base on the pattern, ratio is made. Simple approach easy to use.

- **Ratio Analysis**

This means making the forecast base on the ratio between some casual factors and the number of employees required.

To achieve Rs. 100000 revenue of sales require the 10 employees, that means one employee generates the Rs. 10000 sales

Manning Table / Staffing Table

The short term employee needs can be recorded in a table what is called as “Staffing table” or “ Manning table.”

Tusitha Company Limited

For the next year ending 31st December 2025

Manning Table

Job No.	Job	Department	Quarterly expected job vacancies					Remarks
			Total	First	Second	Third	Fourth	
4211	Executive	IT	3	2	-	1	-	
549	Driver	Administration	4	-	2	-	2	
10cc	Officer	Finance	2	1	1	-	-	
1122	Clerk	Accounting	3	3	-	-	-	
103	Trainee	Administration	4	-	-	-	4	
8020	Peon	Marketing	3	-	-	2	1	

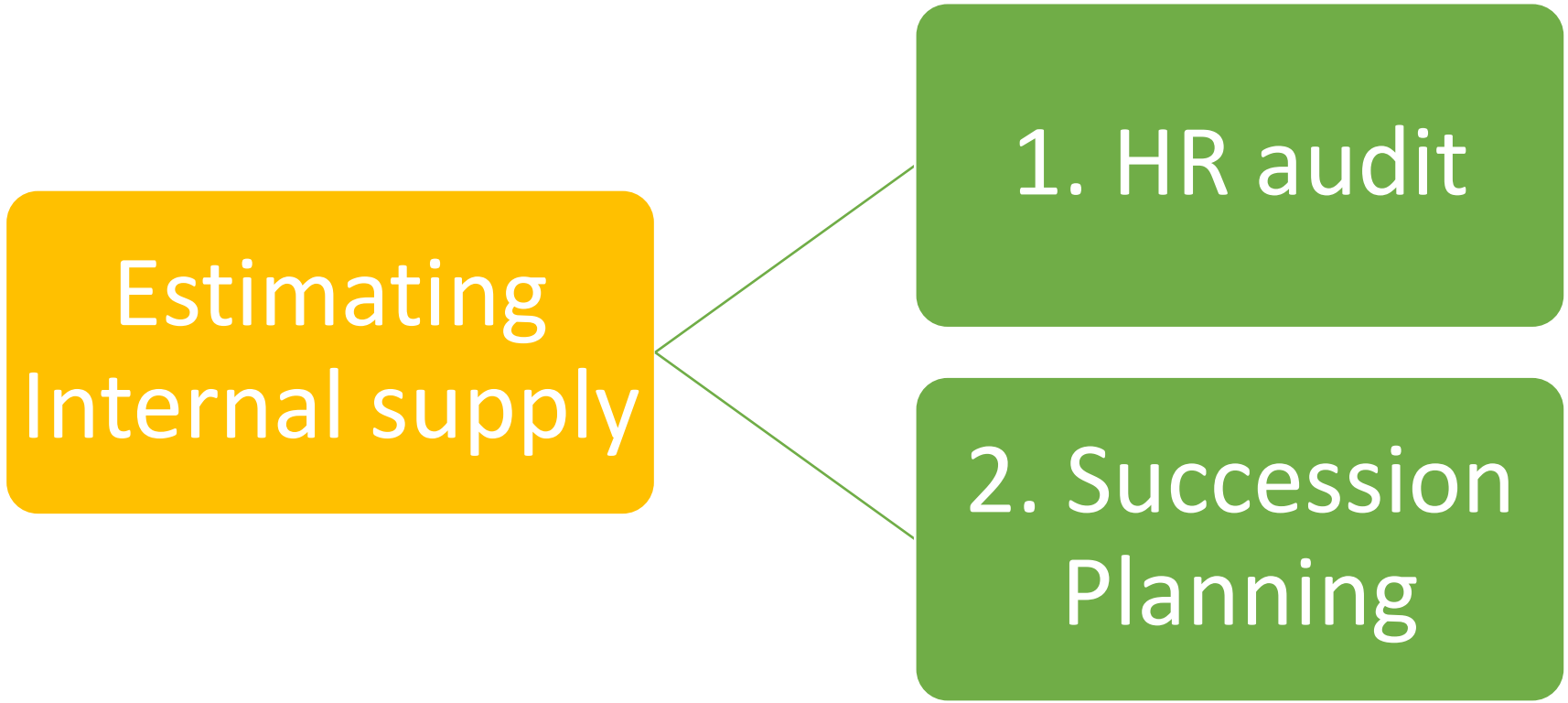
2. Estimates the HR Supply

There are two types of the Human Resource Supply.

1. Internal Supply :- Consists of who can be promoted and transferred to meet forecasted need or fill job vacancies which are in the staffing table.

2. External Supply:- The persons who are working in the other organizations and the people who are seeking the jobs

The internal supply consists of current employee who can be promoted, Transferred or demoted to meet forecast needs. People in the labour market considered under external supply.



A. Auditing the current Human Resource

This means the assessing the current employees' experiences, Competencies and education of the all current employees in the organization.

The HR Audit summarize the all employees' experience, competencies and education.

B. Identifying and Arranging possible Replacement

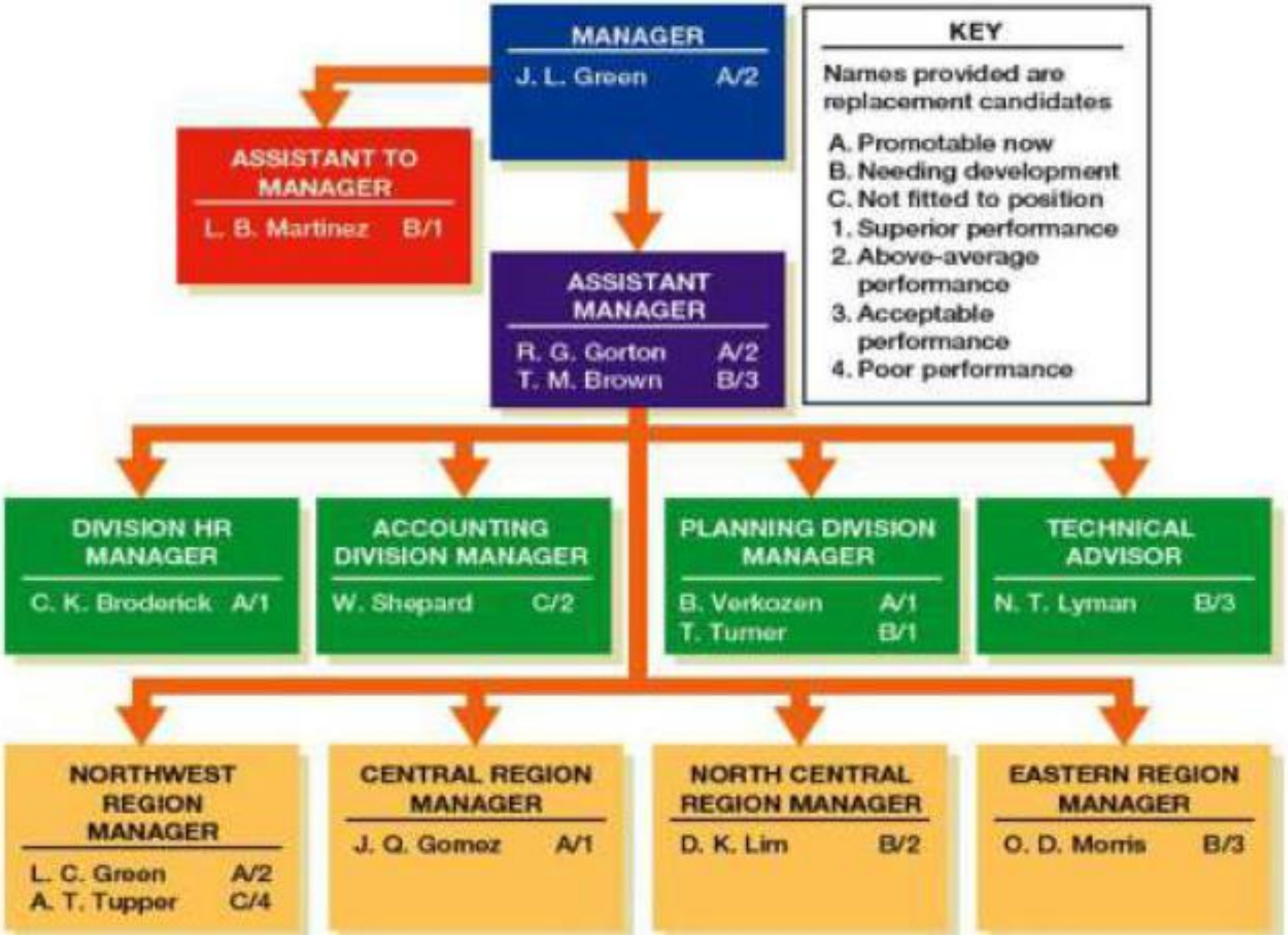
Deciding an employee to fill a job vacancy is a replacement.

Replacements or tentive assignments can be recorded on replacement charts.

The replacement chart included,

- Job holder’s information
- The job’s information
- Their ages
- Current experiences
- Years of experience in service
- Promotional potential
- The successor

A replacement Chart – A visual representation of who will replace whom in the event of the job opening (also called as succession plan).


MANAGER
J. L. Green A/2
ASSISTANT TO MANAGER
L. B. Martinez B/1
ASSISTANT MANAGER
R. G. Gorton A/2
T. M. Brown B/3
KEY
Names provided are replacement candidates
A. Promotable now
B. Needing development
C. Not fitted to position
1. Superior performance
2. Above-average performance
3. Acceptable performance
4. Poor performance
DIVISION HR MANAGER
C. K. Broderick A/1
ACCOUNTING DIVISION MANAGER
W. Shepard C/2
PLANNING DIVISION MANAGER
B. Verkozen A/1
T. Turner B/1
TECHNICAL ADVISOR
N. T. Lyman B/3
NORTHWEST REGION MANAGER
L. C. Green A/2
A. T. Tupper C/4
CENTRAL REGION MANAGER
J. Q. Gomez A/1
NORTH CENTRAL REGION MANAGER
D. K. Lim B/2
EASTERN REGION MANAGER
O. D. Morris B/3

3. Compare HR demand and supply

This involves determining net employee requirements and identifying shortages and surpluses of HR. Identified net requirements of employees can be reported in a specific table called “ **Net employee requirement table**.”

Thusitha Company Limited
For the next year ending 31st December 2025
Net employee requirement table

Employee Category	Quantity of employees		
	Forecasted demand	Estimated supply	Net requirement
Executive	3	2	-1
Driver	4	5	+1
Officer	2	0	-2
Clerk	3	3	0
Trainee	4	2	-2
Total	16	12	-4

4. Decides strategies to be taken

Calculate the net employee requirement of the relevant period of time. The net requirement can be positive or negative one.

If Positive – There is a surplus of the employees. Organization has the employees those are more than the number of employees forecasted for the period

Negative – Shortage, It means the less than the number of employees forecasted for the period

Surplus

- Hiring freeze
- Reduction of the reward expenditure
- Voluntary Departure
- Early retirements
- Lay off
- Encouragement of leave absence
- Termination

Shortage

- Having current employees for the overtime
- Subcontract workers
- Hire part time or fultime employees
- Hire casual employees
- Capital substitution

Assess HRP Effort

Evaluating success or failure of the HRP is a difficult task.

1. Did job vacancies which had not forecasted occur? Not?
2. What is the difference between forecasted number and actual number of vacancies with regard to a job?
3. Are there current employees who are suitable for filling the job vacancies but who were not considered for replacement?
4. Are there any job vacancies which were not filled at the right time during the period being considered?
5. How long job vacancies remain vacant?

Analyzing Hiring Challenges in Hiring Tech Talents

- Shortage of skilled workers in specialized tech roles (e.g., cybersecurity, data science).
  - E.g- A leading cybersecurity firm like **CrowdStrike** faces a **shortage of cybersecurity experts** due to the increasing frequency of cyberattacks.
  - There are more than **3 million unfilled cybersecurity jobs** globally, making it difficult for companies to secure top talent in this area.

Analyzing Hiring Challenges in Hiring Tech Talents

- Intense competition among tech companies to secure top talent, often driving salaries and benefits up.

Companies like **Google**, **Amazon**, and **Apple** are engaged in a fierce battle to attract top-tier talent in fields like **data science** and **machine learning**. For instance, **Amazon** may offer **higher salaries, stock options, and bonuses** to lure data scientists from competitors like **Google**.

Analyzing Hiring Challenges in Hiring Tech Talents

- Geographical constraints: Limited availability of tech talent in certain areas, though remote work can help alleviate this challenge.

Companies like **GitHub** and **Slack** have adopted **remote-first models**, giving them access to talent anywhere in the world.

Analyzing Hiring Challenges in Hiring Tech Talents

- Retention issues: Tech employees often switch companies for better opportunities, making employee retention critical.

Companies like **Netflix** and **Spotify** have implemented strategies to **retain employees**, such as offering **flexible working hours**, **career development opportunities**, and **equity compensation** to keep their workforce engaged and motivated.

Thank you…